PatchTST Discharge Forecasting

Casalecchio di Reno — Station 425
Reno River · Progea / TOPKAPI Integration

Technical Report · July 2026

Val NSE	Test NSE	NSE peak	PBIAS
0.9025	**0.9340**	**0.8555**	**–6.1%**
training	2024 unseen	top 10% flows	slight under
Conf. coverage	Peak coverage	Band width	Max observed
92.8%	**95.6%**	**44.3**	**910.6**
target 90%	n=159 events	m³/s mean	m³/s 18 Oct

1. Project Overview

This project implements an hourly discharge forecasting pipeline for the Reno river at Casalecchio di Reno (ARPA station 425) using the PatchTST transformer architecture with calibrated conformal prediction uncertainty bands. The system is designed for operational integration with the PAB (Piano di Allertamento Bolognese) flood-warning infrastructure, ingesting TOPKAPI hydrological model output and producing probabilistic discharge forecasts with a calibrated 90% prediction band.

The pipeline covers the complete ML workflow: raw data ingestion and feature engineering, model training with flood-aware sample weighting, multi-metric evaluation on a fully held-out 2024 test year, conformal band calibration on a full 2023 validation set, and operational one-step-ahead forecasting.

Station and catchment

Parameter	Value
Station	Casalecchio di Reno — ARPA station 425
River	Reno (Apennine catchment, Emilia-Romagna)
Data file	425.sbs.ts (TOPKAPI .sbs format, 22.7 MB)
Record	2013-01-01 to 2026-01-01 (113,953 hourly rows)
Used range	2014-01-01 to 2024-12-31 after date filter
Features (17)	Rain, Prec, Evap, Snow, Temp, Etp, Soil, SoilSat, Perco, YSnow, SWE + QM lags 1/2/3/7/14/21 h
Target	QM — hourly measured discharge (m³/s)
QM train max	1,249.3 m³/s

2. Data Preparation

The raw TOPKAPI time series is parsed, cleaned, and split into three non-overlapping sets. Fix A uses the full calendar year 2023 as the validation set, ensuring the conformal calibration set contains real autumn (SON) flood events, which are the most hydraulically significant on the Reno. Training samples with QM > 50 m³/s are repeated 8x, growing the training set from 78,807 to 126,815.

Split	Period	Rows	QM mean	QM max	n>200
Train	2014-01-01 to 2022-12-31	78,867	14.8	1,249.3 m³/s	940
Val	2023-01-01 to 2023-12-31	8,760	16.8	648.9 m³/s	102
Test	2024-01-01 to 2024-12-31	8,784	29.0	910.6 m³/s	159

All four seasons are represented in the val set (DJF n=2,160 · MAM n=2,208 · JJA n=2,208 · SON n=2,184), with SON max QM=504.3 m³/s providing critical calibration data for autumn flooding.

3. Model Architecture

The model is a channel-independent PatchTST (Nie et al., 2022) with dual output heads. Each of the 17 input features is processed independently through the same transformer weights. Patches of length 10 h with stride 5 h are extracted from a 60-hour lookback window, yielding 11 patches per feature.

Hyperparameter	Value	Notes
seq_len / patch / stride	60 h / 10 h / 5 h	11 patches per feature
d_model / n_heads / n_layers	128 / 4 / 4	transformer dimensions
d_ff / dropout	512 / 0.10	feedforward width
Parameters	7,037,954	total trainable weights
batch_size / epochs / patience	64 / 400 / 80	flood-weighted sampler
Loss	Huber (delta=5) w1=0.3, w2=1.0	dual-head weighted loss
Optimizer / scheduler	AdamW / OneCycleLR	max_lr=3e-4
Head 1 (log)	log1p(QM) normalised	accurate at low flows
Head 2 (mag)	QM / QM_max (Softplus)	magnitude-preserving for floods
Blend	sigmoid at 15 m³/s, scale 10	smooth transition between heads

The dual-head design addresses the multi-modal nature of Apennine discharge: the log-head captures low-flow dynamics precisely (base flow 5-30 m³/s) while the magnitude head correctly scales during floods up to 910 m³/s.

4. Evaluation Results

The model was evaluated on the completely held-out 2024 test year (8,784 hours), not seen during training or conformal calibration. The year 2024 was an exceptionally challenging test, containing the largest flood on record in the dataset (910.6 m³/s, 18 October) and multiple autumn extreme events.

Metric	Test set (2024)	Full dataset (2014-2024)	Benchmark
NSE	0.9340	0.9782	> 0.80 = excellent
NSE peak	0.8555	0.9628	> 0.70 = good
PBIAS	–6.06%	–3.12%	< ±10% = acceptable
RMSE	16.2 m³/s	7.6 m³/s	—
RMSE peak	49.3 m³/s	22.3 m³/s	threshold >71 m³/s
MAE	4.1 m³/s	1.7 m³/s	—

Test NSE of 0.934 on completely unseen 2024 data is strong for operational hydrological forecasting. NSE peak of 0.856 on the top decile of flows is particularly significant, as flood-peak accuracy is the metric most relevant to the PAB alert system. Best val NSE at training: 0.9025.

5. Conformal Prediction Bands

Uncertainty bands are calibrated using split conformal prediction on the full 2023 validation set (8,700 samples). This is a distribution-free, statistically rigorous method guaranteeing marginal coverage at the target level (90%) with finite-sample validity, without assuming any distributional form for the residuals.

Coverage results

Metric	Before Fix A	After Fix A	Change
Overall coverage	92.4%	92.8%	+0.4 pp
Peak coverage >200 m³/s	43.8%	95.6%	+51.8 pp
Peak events (n)	96	159	+65
SON calibration samples	0 (fallback)	2,184 real	fixed
Mean band width	—	44.3 m³/s	—
Peak q-hat upper	narrow	+249.95 m³/s	correctly sized

The peak coverage jump from 43.8% to 95.6% (+51.8 pp) is the primary validation result. The root cause of the previous failure was the old val set (Jan-Jun 2024 only) containing no SON data, causing a severely undersized upper quantile for autumn flood events. Fix A directly resolved this by providing 2,184 real autumn observations (SON max QM = 504.3 m³/s) for calibration.

Seasonal quantiles (2023 calibration set)

Season	n	q lower	q upper	Notes
DJF (winter)	2,100	–7.9	+5.1 m³/s	wider lower bound
MAM (spring)	2,208	–4.7	+11.0 m³/s	wider upper (snowmelt)
JJA (summer)	2,208	–0.3	+4.5 m³/s	tightest (dry season)

SON (autumn)	2,184	–6.6	+5.4 m^3/s	was 0 before Fix A
Peak (>200)	102	–8.3	+249.95 m^3/s	separate quantile

6. Operational Forecast

Script 5 produces the operational forecast for PAB integration. It supports live mode (single next-timestep forecast) and replay mode (historical rerun for validation). A flood alert is raised when Qfor_high (upper band) crosses 200 m^3/s — conservative by design, firing when flooding is plausible.

Sample output — 2026-01-01 00:00 UTC (low-flow period)

Field	Value	Interpretation
Qfor_mid	11.95 m^3/s	Central forecast
Qfor_low	4.01 m^3/s	Lower bound of 90% band
Qfor_high	19.67 m^3/s	Upper bound of 90% band
Q_TOPKAPI	17.47 m^3/s	Raw TOPKAPI — agrees with model
Head disagree	5.15 m^3/s	Low — both heads agree
Rise (3h)	0.022	No rise signal
Flood alert	No	Qfor_high < 200 m^3/s

Model and TOPKAPI agree closely (11.95 vs 17.47 m^3/s), head disagreement is low, and no alert fires — all physically consistent with January base flow.

7. Pipeline Summary

Script	Purpose	Key outputs
0_check_setup.py	Environment check	Console pass/fail report
1_prepare_data.py	Feature engineering	X_train/val/test.npy, scalers
2_train.py	PatchTST training	best_model.pt, checkpoint.pt
3_evaluate.py	Test evaluation	predictions_test.csv, 7 plots
4_conformal.py	Band calibration	conformal_qhats.json, intervals CSV
5_realtime_infer.py	Operational forecast	forecast_latest.csv

8. Known Limitations

One-step-ahead only. The model forecasts a single hour ahead. Multi-hour lead times would require a sequence-to-sequence output head.

CPU training. Training on CPU takes several hours. A CUDA GPU would reduce this to ~20 minutes.

Flood alert threshold. The 200 m^3/s alert level is provisional. The exact PAB threshold for station 425 should be confirmed with the responsible supervisor before deployment.

TOPKAPI dependency. The model uses TOPKAPI output as input features. Any change to the TOPKAPI configuration requires retraining.

Marginal coverage guarantee. The 90% coverage is marginal (on average over the calibration distribution), not conditional on sub-populations.

October 2024 extreme. The 910.6 m^3/s event is the largest in the dataset by a wide margin. The model captures it within the band but such events have limited training representation.